19/02/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3.00 Hrs.

Test - 10

Topics covered :

Physics : Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor

Chemistry : Amines (Organic Compounds containing Nitrogen), Biomolecules, Polymers, Chemistry in Everyday Life

Botany : Ecosystem, Biodiversity and Conservation, Environmental Issues

Zoology : Biotechnology: Principles and Processes, Biotechnology and its Applications

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. If radius of the first Bohr orbit is r, then the de-Broglie wavelength of electron in 4th orbit is nearly

(1) $2\pi r$ (2) $\frac{\pi r}{2}$

(3) $8\pi r$ (4) $4\pi r$

2. When a hydrogen atom is excited from ground state to first excited state, then its potential energy

(1) Decreases by 20.4 eV

(2) Increases by 20.4 eV

(3) Decreases by 10.2 eV

(4) Increases by 10.2 eV

3. If velocity of an electron in second energy level of H-like atom is v, then velocity of electron in fourth energy level is

(1) $\frac{v}{2}$ (2) $\frac{v}{3}$

(3) $\frac{v}{4}$ (4) v

4. A nucleus of ${}_{84}Po^{210}$ originally at rest emits α particle with speed v_0. The recoil speed of daughter nucleus is

(1) $\frac{4v_0}{214}$ (2) $\frac{4v_0}{206}$

(3) $\frac{v_0}{210}$ (4) $\frac{v_0}{4}$

5. If F_{PP} is the nuclear force between proton-proton, F_{nn} is nuclear force between neutron-neutron and F_{Pn} is the nuclear force between proton-neutron, then

(1) $F_{PP} = F_{Pn} = F_{nn}$

(2) $F_{PP} < F_{Pn} < F_{nn}$

(3) $F_{PP} > F_{Pn} > F_{nn}$

(4) $F_{Pn} < F_{PP} = F_{nn}$

6. The binding energy per nucleon for ${}_1^1H$, ${}_1^3H$ and ${}_2^4He$ are *a*, *b* and *c* respectively, then energy released in a hypothetical nuclear reaction

${}_1^1H + {}_1^3H \rightarrow {}_2^4He + E$, is

(1) $a + 3b - 4c$ (2) $4c - a - 3b$

(3) $4c - a + 3b$ (4) $c - a - b$

7. 1 atomic mass unit is

(1) Equal to $\frac{1}{12}$ (mass of one C^{12} atom)

(2) Equivalent to 931.5 MeV

(3) Equivalent to 149 pJ

(4) All of these

8. Choose the incorrect statement.

(1) A free neutron is unstable

(2) A free neutron has mean life of about 1000 s

(3) A free neutron decays into a proton, an electron and an antineutrino

(4) Neutron is unstable inside the nucleus

9. The mass of proton is 1.0073u and that of neutron is 1.0087u, where u is atomic mass unit. The binding energy of ${}_2^4He$ is (given mass of He nucleus = 4.0015u)

(1) 28.4 eV (2) 28.4 MeV

(3) 56.8 eV (4) 56.8 MeV

10. The mean life of a radioactive material, if its activity drops to $\left(\frac{1}{16}\right)^{th}$ of initial value in 80 years, will be nearly

(1) 20 years

(2) 10 years

(3) 29 years

(4) 14 years

11. Consider a hypothetical nuclear reaction in which the number of alpha particles emitted by a radioactive substance is half the number of β^- particles emitted by it. The resulting daughter is

(1) Isomer of parent (2) Isobar of parent

(3) Isotonic of parent (4) Isotope of parent

12. The energy liberated in the decay

${}_{11}^{22}Na \rightarrow {}_{10}^{22}Ne + e^+ + \nu + Q$, is

(1) $\left[m\left({}_{11}^{22}Na\right) - m\left({}_{10}^{22}Ne\right)\right]c^2$

(2) $\left[m\left({}_{11}^{22}Na\right) - m\left({}_{10}^{22}Ne\right) - 2m_e\right]c^2$

(3) $\left[m\left({}_{11}^{22}Na\right) - m\left({}_{10}^{22}Ne\right) - m_e\right]c^2$

(4) $\left[m\left({}_{11}^{22}Na\right) - m\left({}_{10}^{22}Ne\right) - 10m_e\right]c^2$

13. Choose the correct option.

(1) 1 Ci = 3.7 Bq

(2) 1 Ci = 3.7×10^6 Bq

(3) 1 Ci = 3.7×10^{10} Bq

(4) 1 Bq = 3.7×10^{10} decay/s

14. Which among the following is used as a moderator in nuclear reactor?

(1) Cadmium (2) D_2O

(3) Boron (4) Plutonium

15. Fusion reaction takes place at high temperature because

(1) Kinetic energy is high enough to overcome the coulomb repulsion between nuclei

(2) Molecule breaks up at high temperature

(3) Atoms get ionised at high temperature

(4) Nuclei breaks up at high temperature

16. Two nuclei have their mass numbers in the ratio 1 : 8. The ratio of their nuclear densities would be

(1) 1 : 2 (2) $1 : 2\sqrt{2}$

(3) 1 : 1 (4) 1 : 8

17. In an intrinsic semiconductor, the number of conduction electrons is 2.5×10^{18} in one cubic metre. The total number of current carriers in the same semiconductor of size 1 cm × 1 cm × 1 cm is

(1) 5×10^{18} (2) 5×10^{12}

(3) 2.5×10^{18} (4) 7.5×10^{12}

18. A intrinsic semiconductor have equal electron and hole concentration of 1.5×10^{10} cm^{-3}. Doping by phosphorus, increases electron concentration to 2.25×10^{22} m^{-3}. The hole concentration in doped semiconductor is

(1) 10^{10} m^{-3} (2) 10^{9} m^{-3}

(3) 10^{11} m^{-3} (4) 5×10^{9} m^{-3}

19. In a bipolar junction transistor

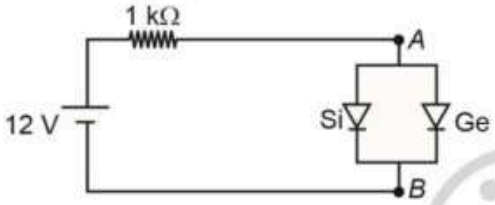
(1) Both emitter and collector are equally doped

(2) Base is heavily doped than collector

(3) Base is lightly doped and thin

(4) Emitter is lightly doped and thin

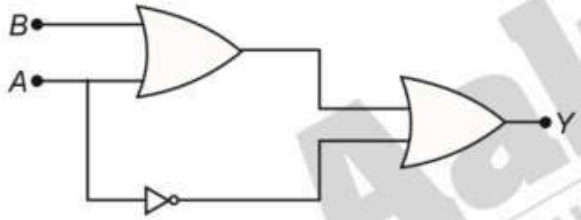
20. A circuit containing diodes is shown in the figure. The potential difference across point A and B ($V_A - V_B$) is



(1) 11.7 V (2) 11.3 V

(3) 0.7 V (4) 0.3 V

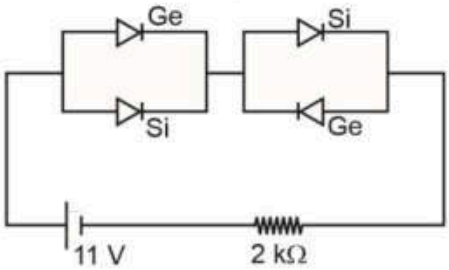
21. A logic circuit is shown in the figure. The correct expression for output Y is



(1) $A + B$ (2) $\bar{A} + B$

(3) 1 (4) B

22. The current flowing through the resistance of 2 kΩ resistance in the given circuit is



(1) 5 mA (2) 5.5 mA

(3) 6 mA (4) Zero

23. An input signal $V = V_0 \sin\omega t$ is applied at input of half wave rectifier. The r.m.s. value of the output for one cycle is

(1) $\frac{V_0}{\sqrt{2}}$ (2) $\frac{V_0}{2}$

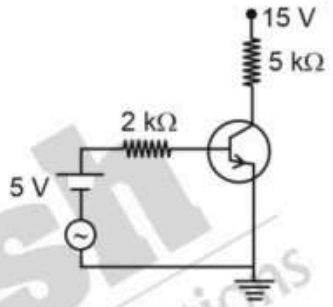
(3) $\frac{2V_0}{\pi}$ (4) $\frac{V_0}{\pi}$

24. For bipolar junction transistor, if $\frac{I_C}{I_E} = x, \frac{I_C}{I_B} = y$ and $\frac{I_E}{I_B} = z$, then incorrect relation among them is

(1) $z = y + 1$ (2) $\frac{1}{x} = \frac{1}{y} + 1$

(3) $y = 1 - \frac{1}{z}$ (4) Both (1) and (2)

25. If the current gain of CE amplifier shown below is β, then transconductance of the amplifier is



(1) $\beta\,\Omega^{-1}$ (2) $\frac{\beta}{2}\,\Omega^{-1}$

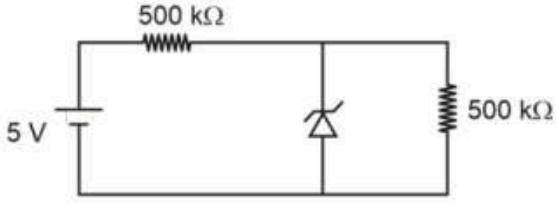
(3) $\frac{\beta}{2} \times 10^{-3}\,\Omega^{-1}$ (4) $\beta \times 10^{-3}\,\Omega^{-1}$

26. A p-n photodiode is manufactured from a semiconductor with E_g = 2.4 eV. Which among the following wavelengths can be detected by it?

(1) 5000 Å (2) 5000 nm

(3) 5500 Å (4) Both (1) and (3)

27. A Zener diode of breakdown voltage V_Z = 10 V is used in voltage regulator circuit as shown in the figure. The current through the diode will be



(1) 5 mA (2) 2.5 mA

(3) 20 mA (4) Zero

28. Among the following, which diode is forward biased?

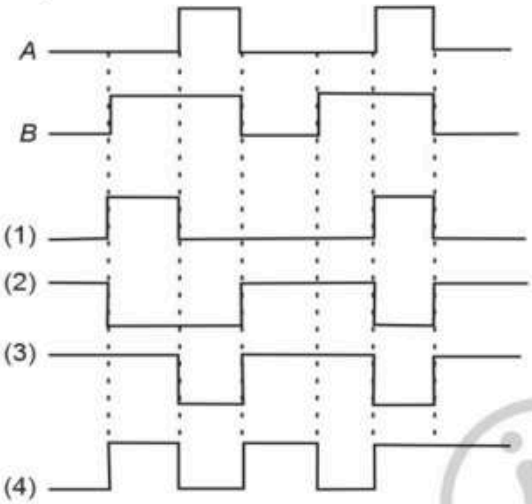
(1) 5 V ⟶|⟶ 10 V

(2) –2 V ⟶|⟶ –1 V

(3) –2 V ⟶|⟶ 0 V

(4) 1 V ⟶|⟶ 4 V

29. The input waveforms are shown below which are applied at the inputs of NAND gate. The correct output wave form will be



30. Choose the incorrect statement.

(1) In forward biased p-n junction, barrier potential decreases

(2) In *CE* amplifier, input and output signals are out of phase

(3) For sustained oscillations the condition $AB = 1$ should be satisfied

(4) Depletion width of p-n junction, increases with increase in doping concentration.

31. If E_1, E_2, E_3 and E_4 are energy of first, second, third and fourth orbit of H-atom respectively, then correct order of energy difference is

(1) $E_2 - E_1 < E_3 - E_2 < E_4 - E_3$

(2) $E_4 - E_3 < E_3 - E_2 < E_2 - E_1$

(3) $E_2 - E_1 = E_3 - E_2 = E_4 - E_3$

(4) $E_3 - E_2 < E_4 - E_3 < E_2 - E_1$

32. The kinetic energy of an electron with de Broglie wavelength of λ is

(1) $\frac{hc}{\lambda}$ (2) $\frac{h^2}{2m\lambda^2}$

(3) $\frac{h}{\sqrt{2m\lambda}}$ (4) $\frac{h^2}{m\lambda^2}$

33. The work function of a metal is 2.5 eV. A light of wavelength 4000 Å is incident on this metal surface. The maximum velocity of emitted photo electrons will be about

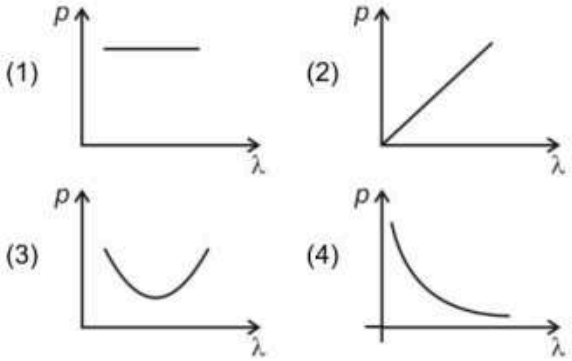
(1) 7.5×10^5 m/s

(2) 9.2×10^5 m/s

(3) 4.6×10^3 m/s

(4) 4.6×10^5 m/s

34. Which of the following figures, represent the variation of particle momentum (p) and the associated de-Broglie wavelength (λ)?



35. The number of photo-electrons emitted for light at a frequency $\nu(\nu > \nu_0)$ is proportional to

(1) Threshold frequency ν_0

(2) $\nu > \nu_0$

(3) Intensity of light

(4) Frequency of light ν

36. Two radiations of photon energies 4.5 eV and 6.5 eV successively illuminate a photosensitive metallic surface of work function 2.5 eV. The ratio of maximum kinetic energies of the emitted electrons is

(1) 1 : 1 (2) 2 : 3

(3) 1 : 2 (4) 1 : 4

37. Which among the following shows the particle nature of light?

(1) Interference

(2) Polarisation

(3) Photoelectric effect

(4) Diffraction

38. If particles are moving with same velocities, then which among the following have maximum de-Broglie wavelength?

(1) Electron (2) Proton

(3) Neutron (4) α-particle

39. When photons of energy $h\nu$ falls on a metal plate of work function ϕ_0, then photoelectrons of maximum kinetic energy K are ejected. If frequency of radiation is doubled, then maximum kinetic energy of ejected photoelectron will be

(1) $2K + \phi_0$ (2) $K + h\nu$

(3) $2K$ (4) Both (1) and (2)

40. A proton is accelerated through 169 V. Its de-Broglie wavelength will be

(1) 2.2 nm (2) 2.2 pm

(3) 94 pm (4) 1.5 pm

41. A radio transmitter operates at a frequency 990 kHz and power of 10 kW. The number of photons emitted per second is

(1) 1.53×10^{31}

(2) 6.6×10^{34}

(3) 1.53×10^{-31}

(4) 8.8×10^{30}

42. A situation of population inversion is related to

(1) Matter wave (2) LASER

(3) LED (4) Photodiode

43. The wave number of the last line of Lyman series in H-spectrum is (Rydberg constant $R = 10^7\ m^{-1}$)

(1) $\frac{3}{4} \times 10^7\ m^{-1}$

(2) $2 \times 10^7\ m^{-1}$

(3) $10^7\ m^{-1}$

(4) $4 \times 10^7\ m^{-1}$

44. When a hydrogen atom is raised from first excited state to 2nd excited state, then

(1) Both KE and PE increases

(2) Both KE and PE decreases

(3) PE increases and KE decreases

(4) KE increases and PE decreases

45. Assuming Bohr's model for Li^{++}, the first excitation energy is

(1) 17 eV

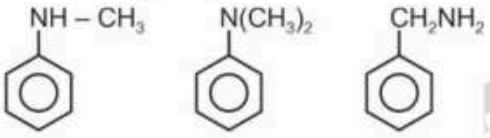
(2) 1.89 eV

(3) 7.56 eV

(4) 91.8 eV

CHEMISTRY

46. The correct decreasing order of basic strength for the following compounds is



(I) (II) (III)

(1) II > I > III (2) III > I > II

(3) II > III > I (4) III > II > I

47. Nitrobenzene on reaction with Zn dust in the presence of aq NaOH gives

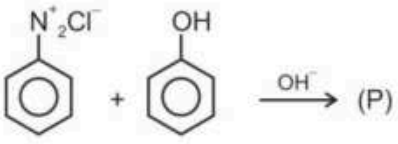
(1) Hydrazobenzene

(2) Azobenzene

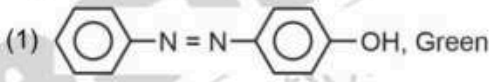
(3) Azoxybenene

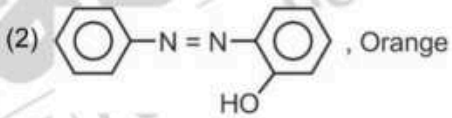
(4) Aniline

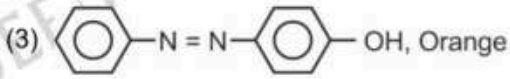
48. Consider the following reaction,

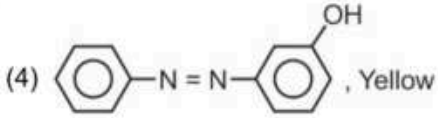


The compound P and its colour respectively are

(1) 

(2) 

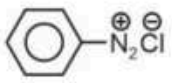
(3) 

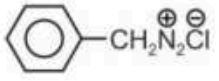
(4) 

49. The most stable diazonium salt among the following is

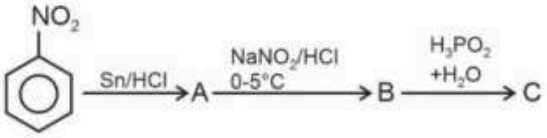
(1) $CH_3\overset{\oplus}{N_2}Cl^{\ominus}$

(2) $(CH_3)_2CH\overset{\oplus}{N_2}Cl^{\ominus}$

(3) 

(4) 

50. Consider the following reaction sequence,



C is

(1) 

(2) 

(3) 

(4) 

51. In which of the following reactions, primary amines are formed?

I. $R-CN \xrightarrow{Na(Hg)/C_2H_5OH}$

II. $R-CO-NH_2 \xrightarrow[(ii)\,H_2O]{(i)\,LiAlH_4}$

III. $R-NC \xrightarrow[(ii)\,H_2O]{(i)\,LiAlH_4} A$

(1) I only
(2) I and II only
(3) II and III only
(4) I, II and III

52. Hinsberg reagent is used to distinguish between

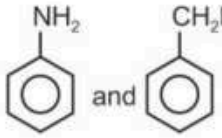
(1) 

and CH_3NH_2

(2) 

and $CH_3CH_2NH_2$

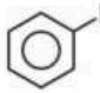
(3) CH_3NH_2 and CH_3NHCH_3

(4) 

53. Gabriel phthalimide synthesis is used to prepare

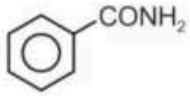
(1) Aromatic primary amines
(2) Aliphatic secondary amines
(3) Aliphatic tertiary amines
(4) Aliphatic primary amines

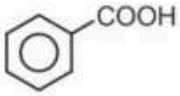
54. Consider the following reaction,

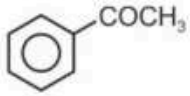
$A + Br_2 + NaOH \longrightarrow$ 

$+ Na_2CO_3 + NaBr + H_2O$

A in the above reaction is

(1) 

(2) 

(3) 

(4) 

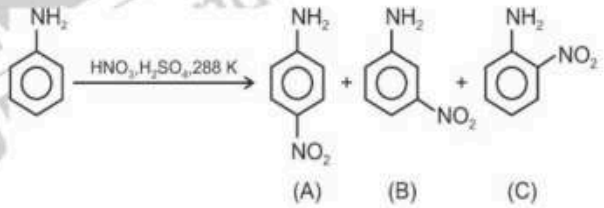
55. Correct order of boiling point of isomeric amines is

(1) Tertiary > Primary > Secondary
(2) Secondary > Tertiary > Primary
(3) Primary > Secondary > Tertiary
(4) Tertiary > Secondary > Primary

56. The correct IUPAC name of the following compound $C_2H_5-\underset{\displaystyle C_2H_5}{\overset{|}{N}}-C_4H_9$ is

(1) N-Butylmethanamine
(2) N-Butyl-N-ethylethan-1-amine
(3) N, N-Diethylbutan-1-amine
(4) N-Ethyl-N-butylethan-1-amine

57. Consider the following reaction,



The correct order of yield of the products obtained is

(1) A > C > B
(2) C > A > B
(3) B > A > C
(4) A > B > C

58. Lactose is composed of

(1) β-D-glucose and β-D-fructose
(2) β-D-galactose and β-D-glucose
(3) α-D-glucose and β-D- galactose
(4) α-D-glucose and β-D-glucose

59. Glucose on oxidation with nitric acid gives
(1) n-hexane (2) Saccharic acid
(3) Gluconic acid (4) Adipic acid

60. Non-reducing sugar among the following is
(1) Glucose (2) Maltose
(3) Sucrose (4) Fructose

61. Identify the incorrect statement.
(1) Amylose is water soluble
(2) Amylose is a long unbranched chain
(3) Amylopectin constitutes about 80-85% of starch
(4) Amylopectin is a branched chain polymer of β-D-glucose units

62. Water soluble vitamin among the following is
(1) Vitamin C (2) Vitamin A
(3) Vitamin D (4) Vitamin K

63. The nitrogenous base which is not present in DNA is
(1) Adenine (2) Guanine
(3) Uracil (4) Cytosine

64. Basic amino acid among the following is
(1) Proline (2) Leucine
(3) Serine (4) Lysine

65. Rickets is caused by the deficiency of
(1) Vitamin B_{12} (2) Vitamin D
(3) Vitamin A (4) Vitamin B_6

66. Pentose sugar among the following is
(1) Ribose (2) Galactose
(3) Mannose (4) Fructose

67. Which of the following structure(s) of proteins is/are destroyed during denaturation of proteins?
(1) Primary only
(2) Primary and secondary only
(3) Secondary and tertiary only
(4) Primary, secondary and tertiary

68. Which of the following is fat-soluble vitamin?
(1) Retinol
(2) Thiamine
(3) Ascorbic acid
(4) Riboflavin

69. Nucleotides are joined together by
(1) Peptide linkage
(2) Phosphodiester linkage
(3) Glycosidic linkage
(4) Ether linkage

70. Six membered cyclic structure of glucose is named as glucopyranose with analogy to the compound
(1) Pyrrole (2) Pyridine
(3) Furan (4) Pyran

71. The enzyme that catalyses the acid hydrolysis of sucrose is
(1) Zymase (2) Maltase
(3) Sucrase (4) Pepsin

72. α-D-glucopyranose and β-D-glucopyranose differs only in the configuration of the hydroxyl group at
(1) C1 (2) C5
(3) C3 (4) C4

73. Which of the following react with glucose and confirm the presence of carbonyl group?

$(CH_3CO)_2O$ HNO_3 NH_2OH HCN
I II III IV

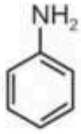
(1) I only
(2) I and III only
(3) III and IV only
(4) I, II, III, and IV

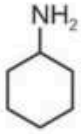
74. Incorrect statement about cellulose is
(1) Predominant constituent of cell wall of plant cells
(2) Straight chain polysaccharide
(3) Composed of β-D-glucose units
(4) Glycosidic linkage exist between C-1 of one glucose unit and C-6 of the next glucose unit

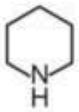
75. Biodegradable polymer which can be produced from 3-hydroxybutanoic acid and 3-hydroxypentanoic acid is
(1) PHBV
(2) Nylon-2-Nylon-6
(3) Nylon-6
(4) Nylon 6, 6

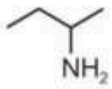
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76. The compound which does not give carbylamine reaction is

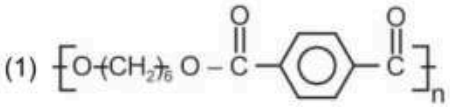
(1) 

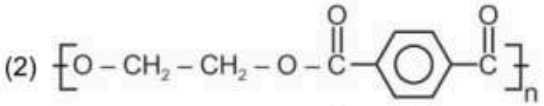
(2) 

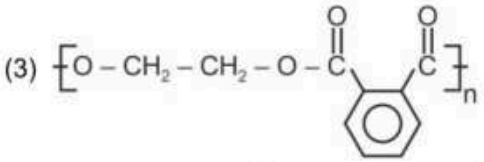
(3) 

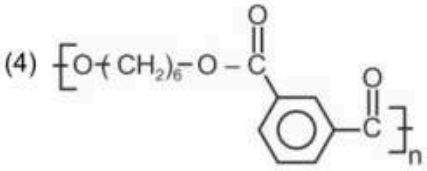
(4) 

77. Which one of the following structures represents Dacron polymer?

(1) 

(2) 

(3) 

(4) 

78. Caprolactam is a monomer of
(1) Nylon 6
(2) Nylon-2-Nylon-6
(3) Polyacrylonitrile
(4) Neoprene

79. Which of the following is a copolymer?
(1) Polystyrene (2) Buna-S
(3) PVC (4) Polythene

80. Which of the following polymers is a polyester?
(1) Urea formaldehyde resin
(2) Nylon-6,6
(3) Glyptal
(4) Buna-N

81. Orlon is a polymer of
(1) $CF_2 = CF_2$ (2) $CCl_2 = CH_2$
(3) $CH_2 = CH_2$ (4) $CH_2 = \underset{|}{\overset{CN}{C}}H$

82. Match the polymers given in column I with their uses given in column II and identify the correct code given below.

	Column I		Column II
(P)	Urea-formaldehyde Resin	(a)	Wrapping material
(Q)	Glyptal	(b)	Manufacture of unbreakable cups and laminated sheets
(R)	Bakelite	(c)	Manufacture of paints and lacquers
(S)	Polystyrene	(d)	Manufacture of electrical switches and handles of utensils

(1) (P)-(a), (Q)-(b), (R)-(c), (S)-(d)
(2) (P)-(a), (Q)-(c), (R)-(d), (S)-(b)
(3) (P)-(b), (Q)-(c), (R)-(a), (S)-(d)
(4) (P)-(b), (Q)-(c), (R)-(d), (S)-(a)

83. Natural Rubber is
(1) cis-1,4-polychloroprene
(2) trans-1,4-polyisoprene
(3) cis-1,4-polyisoprene
(4) trans-1,4-polychloroprene

84. Ortho-sulphobenzimide is also known as
(1) Saccharin (2) Aspartame
(3) Sucralose (4) Alitame

85. Which of the following is an antioxidant?
(1) Sodium benzoate (2) BHT
(3) Vegetable oil (4) Table salt

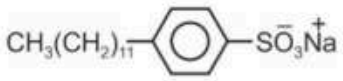
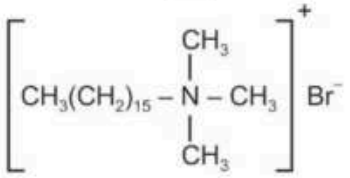
86. Which among the following is not a broad spectrum antibiotic?
(1) Chloramphenicol
(2) Vancomycin
(3) Ofloxacin
(4) Penicillin G

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87. Tranquilizer among the following is
(1) Prontosil (2) Veronal
(3) Dimetapp (4) Seldane

88. Norethindrone is an
(1) Antiseptic (2) Antifertility drug
(3) Antihistamine (4) Analgesic

89. 0.2 percent solution of phenol is a/an
(1) Antacid (2) Antiseptic
(3) Antibiotic (4) Disinfectant

90. Cationic detergent among the following is
(1) $CH_3(CH_2)_{10}CH_2OS\overset{-}{O}_3\overset{+}{N}a$
(2) 
(3) 
(4) $CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$

BOTANY

91. Which of the following may be considered as man-made ecosystem?
(1) Pond and lakes
(2) Wetland and estuary
(3) Crop-fields and aquarium
(4) Desert and grassland

92. Vertical distribution of different species occupying different levels is called
(1) Stratification (2) Scarification
(3) Standing crop (4) Biosphere

93. The rate of formation of new organic matter by the consumers of the ecosystem is known as
(1) Gross primary productivity
(2) Net primary productivity
(3) Secondary productivity
(4) Secondary carnivores

94. Which of the following is the main reason for the low productivity of oceans?
(1) Absence of carbon
(2) Excess of nitrogen
(3) Light decreases with depth
(4) Excess availability of PAR

95. Select the **incorrect** statement w.r.t. humus.
(1) Being colloidal in nature it serves as a reservoir of nutrients
(2) Dark-coloured amorphous substance
(3) It is highly resistant to microbial action
(4) It undergoes decomposition at an extremely fast rate

96. The rate of decomposition is controlled by chemical composition of detritus and climatic factors. Decomposition rate is slower if
(1) Detritus is rich in lignin, chitin and the temperature is low along with anaerobiosis
(2) Detritus is rich in nitrogen, sugar and atmosphere is warm and moist
(3) Conditions are aerobic with nitrogen and sugar rich detritus
(4) Warm and moist conditions nitrogen rich detritus and aerobic conditions are present

97. Select the **odd** statement w.r.t., detritus food chain (DFC).
(1) DFC begins with dead organic matter
(2) It is made up of decomposers mainly fungi and bacteria
(3) Major conduit for energy flow in an aquatic ecosystem
(4) Energy and nutrient requirements are provided by degrading dead organic matter or detritus

98. The mass of living material present at each trophic level at a particular time is called
(1) Standing crop (2) Standing state
(3) Food web (4) Ecosystem

99. Select the **incorrect** statement w.r.t. ecological pyramids.
(1) The pyramid of biomass in sea is generally inverted
(2) Pyramid of energy is always upright and can never be inverted
(3) Saprophytes are given important place in ecological pyramids as they play a vital role in the ecosystem
(4) It does not take into account the same species belonging to two or more trophic levels

100. Secondary succession begins in areas where natural biotic communities have been destroyed such as in

(1) Newly created reservoir

(2) Newly cooled lava

(3) Bare rock

(4) Abandoned farm lands

101. Select the **odd** statement w.r.t. ecological succession?

(1) As succession proceeds, the numbers and types of animals and decomposers also change

(2) The species that invade a bare area are called the seres

(3) All succession whether taking place in water or on land, proceeds to a similar climax community- the mesic

(4) The individual transitional communities are termed as seral stages or seral communities

102. Match the terms given in column-I, with their true explanation given in column-II, and select the **correct** option.

	Column-I		**Column-II**
a.	Primary succession	(i)	Amount of nutrient present in soil
b.	Standing state	(ii)	Is in near equilibrium with the environment
c.	Climax community	(iii)	Degradation of humus by microbes
d.	Mineralisation	(iv)	Where no living organism ever existed

(1) a(ii), b(iii), c(iv), d(i)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(i), b(ii), c(iii), d(iv)

103. Select the **incorrect** statement w.r.t. nutrient cycle.

(1) Nutrient cycles are of two types, gaseous and sedimentary

(2) The reservoir for gaseous type of nutrient cycle is located in earth's crust

(3) Environmental factors like soil, moisture, pH, temperature etc regulate the rate of release of nutrients into atmosphere

(4) The function of the reservoir is to meet with the deficit which occurs due to imbalance in the rate of influx and efflux

104. Which of the following is the major and important difference between carbon and phosphorous cycle?

(1) Atmospheric inputs of phosphorus through rainfall are much smaller than carbon inputs

(2) Gaseous exchanges of carbon between organisms and environment are negligible

(3) Unlike phosphorus cycle, there is no respiratory release of carbon into atmosphere

(4) Phosphorus constitutes 49 percent of dry weight of organisms and is next only to water

105. Out of the total cost of various ecosystem services, contributions of services like recreation and nutrient cycling accounts for

(1) About 50 percent

(2) About 20 percent

(3) Less than 10 percent

(4) More than 50 percent

106. The variation shown by the medicinal plant *Rauwolfia vomitoria* growing in different Himalayan ranges in terms of potency and concentration of the active chemical reserpine. This is an example of

(1) Genetic diversity

(2) Species diversity

(3) Community diversity

(4) Ecological diversity

107. According to the International Union for Conservation of Nature and natural resources (IUCN) 2004, the total number of plant and animal species described so far is slightly more than

(1) 7 million (2) 20 to 50 million

(3) 1.5 million (4) 3.5 million

108. Among animals, the most species rich taxonomic group, making up more than 70 percent of the total are
(1) Molluscs (2) Fishes
(3) Insects (4) Reptiles

109. If we accept Robert May's global species diversity estimates what is the percentage of total species that have been recorded so far?
(1) 22 percent (2) 44 percent
(3) 30 percent (4) 15 percent

110. Select the **mismatch** w.r.t. country and species diversity of birds :

Country	**: No. of bird species**
(1) Colombia	: 1,300 species of birds
(2) New York	: 105 species of birds
(3) India	: 1,200 species of birds
(4) Greenland	: 56 species of birds

111. The largely tropical Amazonian rain forest in South America is home to around how many different species of plants?
(1) 1,25,000 species
(2) Two million species
(3) More than 40,000 species
(4) 427 species

112. While exploring species-area relationships Alexander von Humboldt observed that
(1) Tropical environments are less seasonal
(2) There is more solar energy available in tropics
(3) There is higher productivity at tropics
(4) Within a region, species richness increased with increasing explored area, but only up to a limit

113. A stable community or ecosystem
(1) Shows too much variation in productivity from year to year
(2) Can be replaced easily by other community
(3) Must be either resistant or resilient to occasional disturbances (natural or man-made)
(4) Show large fluctuations in productivity occasionally

114. In the 'rivet popper hypothesis' Stanford ecologist Paul Ehrlich, used the analogy of loss of rivets on the wings of airplane, with the loss of which type of species in the ecosystem?
(1) Key species
(2) Endangered species
(3) Extinct species
(4) Dominant species

115. In which of the following aspects, the 'Sixth Extinction' presently in progress, is different from the previous episodes?
(1) Human beings are not involved in this recent sixth extinction
(2) It is 100 to 1000 times slower than earlier episodes
(3) Human activities are responsible for the faster rates in the sixth extinction
(4) It is just like the previous episodes and mainly due to natural phenomena

116. Loss of biodiversity in a region may lead to increased
(1) Plant diversity and production
(2) Higher resistance to environmental perturbations like drought
(3) Variability in plant productivity, water use and pest cycle
(4) Stability in ecosystem processes like water use, pest and disease cycles and plant productivity

117. The most important cause driving animals and plants to extinction is
(1) Habitat loss and fragmentation
(2) Over-exploitation
(3) Alien species invasions
(4) Co-extinctions

118. The ethical argument for conserving biodiversity relates to
(1) The aesthetic pleasures we derive from nature
(2) The various ecosystem services provided by nature
(3) The countless direct economic benefits from nature
(4) Moral duty to care for the well-being of other species and pass on our biological legacy to future generations

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119. Select the **odd** one out w.r.t. the various on site conservation strategies.
(1) Sacred groves
(2) Biosphere reserves
(3) Wildlife safari parks
(4) National parks

120. The historic convention on Biological Diversity which called upon all nations to take appropriate measures for conservation of biodiversity and sustainable utilisation of its benefits was held in
(1) 1992 at Rio de Janeiro
(2) 2002 at Johannesburg
(3) 2010 at Johannesburg
(4) 1989 at Montreal

121. Which of the following can remove over 99 percent particulate matter present in the exhaust from a thermal power plant?
(1) Scrubber
(2) Catalytic converters
(3) Electrostatic precipitator
(4) Distillation

122. When was the Air (Prevention and control of pollution) Act amended to include noise as an air-pollutant?
(1) 1981
(2) 1987
(3) 1986
(4) 1989

123. When the exhaust passes through the catalytic converter, unburnt hydrocarbons are converted into
(1) CO and nitrogen gas
(2) Nitric oxide
(3) Carbon dioxide and water
(4) Carbon monoxide and H_2S

124. Euro-III norms stipulate that sulphur and aromatic hydrocarbon should be controlled in petrol at
(1) 350 ppm and 42 percent respectively
(2) 150 ppm and 42 percent respectively
(3) 50 ppm and 42 percent respectively
(4) 350 ppm and 35 percent respectively

125. Select the **incorrect** statement w.r.t. air-pollution.
(1) CNG burns more efficiently unlike petrol or diesel
(2) Use of low-sulphur petrol and diesel will reduce vehicular pollution
(3) Leaded petrol is preferred for catalytic converters in automobiles
(4) PM 2.5 are fine particulates that can be inhaled deep into the lungs and can cause damage to lungs and premature deaths

126. Increase in concentration of a toxicant at successive trophic levels, as it is neither metabolised or excreted is called
(1) Biomagnification (2) Bioprospecting
(3) Biofortification (4) Bioremediation

127. Select the **odd** statement w.r.t. accelerated eutrophication.
(1) Lakes in the past century have been severely eutrophied by sewage and industrial wastes
(2) The prime contaminants are nitrates and phosphates which act as plant nutrients
(3) Pollutants overstimulate the growth of algae, causing a dip in dissolved oxygen vital to other aquatic life
(4) The natural aging of a lake may span thousands of years

128. 'Eco-San' toilets used in many areas of Kerala and Sri Lanka are
(1) Unhygienic
(2) Very expensive
(3) Using lot of water
(4) Practical solution to human waste disposal

129. Most effective and eco-friendly method for the disposal of plastic waste is
(1) Incineration (2) Sanitary landfills
(3) Polyblend (4) Landfill dumping

130. Select the **incorrect** match.

(1)	Hospital waste	:	Incineration
(2)	e-Wastes	:	Recycling
(3)	Open-burning dumps	:	Best for solid waste disposal
(4)	*Eichhornia*	:	Most problematic aquatic weed

131. Which of the following greenhouse gases has the maximum relative contribution to global warming?
(1) Methane
(2) Carbon dioxide
(3) Oxides of nitrogen
(4) Chloro-fluoro-carbon

132. Montreal protocol to control the emission of ozone depleting substances became effective in
(1) 1987 (2) 1989
(3) 1992 (4) 2002

133. The national forest policy (1988) of India has recommended A percent forest cover for the plains and B percent for the hills.
(1) A = 67%, B = 33%
(2) A = 21.54%, B = 33%
(3) A = 33%, B = 67%
(4) A = 33%, B = 21.54%

134. Realising the significance of participation of local communities, the Government of India has introduced in 1980s the concept of
(1) Jhum cultivation
(2) Chipko movement
(3) Soil erosion and desertification
(4) Joint forest management

135. Select the **odd** statement w.r.t. good ozone.
(1) Found in the upper part of atmosphere called stratosphere
(2) Acts as a shield absorbing UV-radiations
(3) It is continuously formed by the action of UV rays on molecular oxygen and also degraded into molecular oxygen in stratosphere
(4) Harms plants and animal in troposphere

ZOOLOGY

136. Who developed a method of removing plasmids from the cell and then reinserting them in other cells?
(1) Herbert Boyer (2) Stanley Cohen
(3) Kary Mullis (4) Werner Arber

137. How many techniques mentioned below are included under biotechnology?
(a) *In vitro* fertilisation leading to a test tube baby.
(b) Synthesising a gene and using it.
(c) Developing a DNA vaccine.
(d) Correcting a defective gene.
(1) Zero (2) Two
(3) Three (4) Four

138. First artificial recombinant DNA, constructed by Stanley Cohen and Herbert Boyer, was introduced in
(1) *Salmonella typhimurium*
(2) *Escherichia coli*
(3) Plasmid of *Salmonella typhimurium*
(4) Antibiotic resistant gene of *E. coli* plasmid

139. The first type II restriction endonuclease was isolated from the bacterium
(1) *Escherichia coli*
(2) *Haemophilus influenzae*
(3) *Entamoeba coli*
(4) *Salmonella typhimurium*

140. Choose the **odd** one w.r.t. restriction endonucleases.
(1) Functions by inspecting the length of a DNA sequence
(2) Cut both the strands of the DNA at specific points
(3) Bind to the DNA and cut the bond between sugar and nitrogenous base
(4) They recognises a specific palindromic nucleotide sequence in the DNA

141. The sticky ends of DNA fragments facilitates the action of which of the given enzyme?
(1) DNA polymerase
(2) DNA ligase
(3) Alkaline phosphatase
(4) Restriction endonuclease

142. Read the given statements and choose the **correct** option.

Statement-A : Unless the same restriction enzyme cuts the vector and the source DNA, the recombinant / hybrid molecule cannot be created.

Statement-B : When DNA, irrespective of source is cut by the same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends.

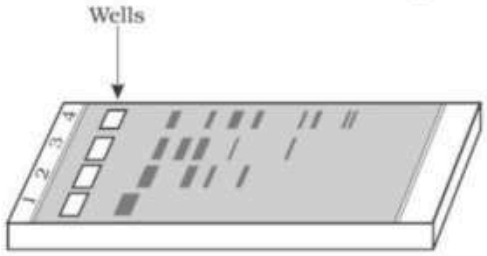
(1) Both statements are incorrect

(2) Both statements are correct

(3) Only statement A is correct

(4) Only statement B is correct

143. The diagram given below shows a typical agarose gel electrophoresis showing migration of undigested (Lane 1) and digested set of DNA fragments (Lane 2 to 4). Smallest DNA fragments are present in lane ____ and are nearer to ____. Choose the option which fill the blanks **correctly**.



(1) 1, cathode (2) 4, cathode

(3) 3, anode (4) 4, anode

144. Choose the **odd** one w.r.t. most commonly used matrix in gel electrophoresis for separation of DNA fragments

(1) It is a natural monomer

(2) Extracted from sea weeds

(3) Provide sieving effect

(4) It is prepared from agar

145. Match the columns and choose the **correct** option.

	Column-I		**Column-II**
a.	Elution	(i)	Ethidium bromide
b.	Gel electrophoresis	(ii)	Chilled ethanol
c.	DNA stain	(iii)	Separation of DNA fragments
d.	Precipitation of DNA	(iv)	Extraction of DNA fragment from the gel piece

(1) a(iv), b(iii), c(i), d(ii)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(iv), c(ii), d(i)

146. How many features given in the box are required in vector to facilitate cloning of alien DNA into it?

Ori site, Selectable marker gene, Cloning sites, Large size of the vector

(1) Four (2) Three

(3) Two (4) One

147. If we ligate alien piece of DNA in pBR322 at the site recognised by *Eco*RI, the resultant recombinants will show

(a) Resistance to ampicillin

(b) Sensitivity to ampicillin

(c) Resistance to tetracycline

(d) Sensitivity to tetracycline

(1) (a) and (c) (2) (b) and (d)

(3) (a) and (d) (4) (b) and (c)

148. How many copies of DNA will form from a dsDNA after 5 cycles of PCR?

(1) 5 (2) 10

(3) 16 (4) 32

149. In PCR, *Taq* polymerase add nucleotide at which end of primer?

(1) 3′-OH end of RNA primer

(2) 5′-P end of RNA primer

(3) 3′-OH end of DNA primer

(4) 5′-P end of DNA primer

150. Primers bind to which end of the DNA strand in PCR?

(1) 3′ end on one strand and 5′ end on another strand

(2) 3′ end on both strands

(3) 5′ end on both strands

(4) Primer binds to 5′ end of one strand only

151. dNTP stands for

(1) Deoxyribonucleotide triphosphate

(2) Deoxyribonucleoside triphosphate

(3) Deoxyribonucleotide monophosphate

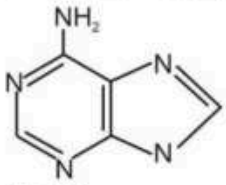
(4) Deoxyribonucleoside monophosphate

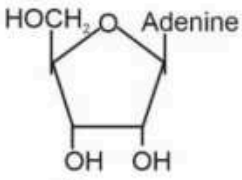
152. *Taq* polymerase is obtained from
(1) Thermophilic bacterium, *Thermus aquaticus*
(2) Thermophilic bacterium, *Bacillus thuringiensis*
(3) Thermolabile bacterium, *Agrobacterium tumefaciens*
(4) Thermolabile bacterium, *Thermus aquaticus*

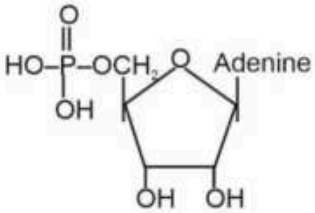
153. The advantage of sparged stirred-tank bioreactor over simple stirred – tank bioreactor is
(1) The stirrer facilitates even mixing and oxygen availability
(2) Presence of O_2 delivery system
(3) Bubbles from sparger dramatically increase the oxygen transfer area
(4) Anaerobic fermentation

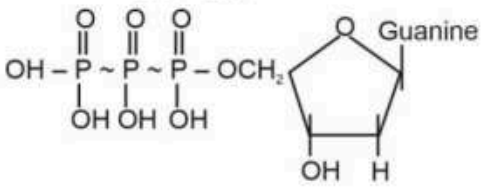
154. Which of the following is **not** a part of downstream processing?
(1) Separation of desired product by processes like filtration, centrifugation etc.
(2) Purification of desired product
(3) Large scale culture of host cells in bioreactors
(4) Use of chromatography for separation of certain products

155. Select the structure of a compound which is used as raw material to carry PCR.

(1) 

(2) 

(3) 

(4) 

156. Which of the below mentioned vectors can be used to transfer gene in a bacteria?
(a) pBR322
(b) Bacteriophage
(c) Phagemid
(d) Cosmid
(1) (a) only (2) (a) and (b) only
(3) (c) and (d) only (4) (a), (b), (c) and (d)

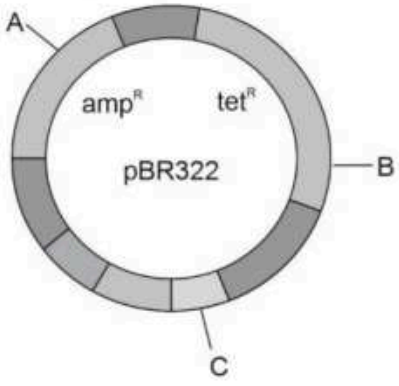
157. Complete the analogy :

Agrobacterium tumefaciens	:	Dicot plants	:	Retro viruses	:	_____

(1) Bacteria
(2) Monocot plants
(3) Fungi
(4) Animal cells

158. If we introduce a alien DNA in the coding sequence of β-galactosidase in pUC8, then what will **not** occur?
(1) Non-transformants will grow on ampicillin containing medium
(2) Transformants will grow on ampicillin containing medium
(3) Recombinants will produce white coloured colonies
(4) Non-recombinants will produce blue coloured colonies

159. Given below is the diagram of cloning vector pBR322. Choose the option which **correctly** represents A, B and C w.r.t. restriction sites.



	A	B	C
(1)	*Pvu* I	*Sal* I	rop
(2)	*Pvu* II	*Sma* I	Ori
(3)	*Pst* I	*Sal* I	*Pvu* II
(4)	*Eco*RI	*Bam*H I	rop

160. Select the **correct** sequence of the steps involved in the process of recombinant DNA technology.

(a) Culturing the host cells at large scale

(b) Isolation of a desired DNA fragment

(c) Transferring the recombinant DNA into the host.

(d) Ligation of the DNA fragment into a vector

(1) (b) → (d) → (c) → (a)

(2) (c) → (d) → (a) → (b)

(3) (a) → (b) → (c) → (d)

(4) (d) → (a) → (b) → (c)

161. Select the **mismatch**.

(1)	Gel electrophoresis	–	Employed to check the progression of a restriction enzyme digestion
(2)	Recombinant protein	–	Protein encoding gene expressed in a homologous host
(3)	Transformation	–	Piece of DNA is introduced in a host bacterium
(4)	Bioprocess engineering	–	Maintenance of sterile ambience

162. Match the following columns and choose the **correct** option.

	Column-I		**Column-II**
a.	Recombinant DNA technology	(i)	More than 900 have been isolated from over 230 strains of bacteria
b.	Traditional hybridisation	(ii)	100-1000 litres of culture can be processed
c.	Bioreactors	(iii)	Early diagnosis of diseases
d.	Restriction enzymes	(iv)	Very often include inclusion of undesirable genes

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

163. Read the following statements and choose the **correct** option.

Statement A : The normal *E.coli* cells do not carry resistance against kanamycin and tetracycline.

Statement B : Selectable marker gene helps in the selection of transformants.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both A and B statements are correct

(4) Both A and B statements are incorrect

164. Select the **correct** match.

(1)	Ori	–	Codes for the proteins involved in the replication of the plasmid
(2)	rop	–	Replication of DNA starts from this sequence
(3)	Palindromic nucleotide sequence	–	Base pairs that reads same on the two strands of DNA when orientation of reading is kept different
(4)	Restriction endonuclease	–	Cut between the same two bases on the opposite strands of DNA

165. Read the following statement and choose the option which **correctly** represents the true (T) and false (F) statements.

A. EFB stands for European Federation of Biotechnology.

B. First restriction endonuclease was *Hind* II

C. Methylase enzyme protect the bacterial DNA from the action of restriction endonuclease.

D. In *Eco*RI, R stands for Rd strain of *E. coli*

	A	B	C	D
(1)	T	T	T	T
(2)	T	T	T	F
(3)	T	T	F	F
(4)	F	T	F	F

166. How many applications given below are of biotechnology?

(a) Bioremediation (b) Waste treatment

(c) Energy production (d) Diagnostics

(1) Two

(2) Three

(3) Four

(4) One

167. The insect which is killed by the proteins encoded by *cry IAc* and *cry II Ab* belongs to which order?
(1) Lepidoptera (2) Coleoptera
(3) Diptera (4) Hymenoptera

168. Why is the gene coding for Bt toxin called 'cry'?
(1) *cry* gene codes for crystalline protein
(2) *cry* gene exist as crystal
(3) *cry* gene codes for cryophilic protein
(4) *cry* gene codes for Bt toxin

169. Select the **correct** match.

(1)	Tailor-made plants	–	Supply alternative resources to industries
(2)	Green revolution	–	Use of organic fertilisers and pesticides
(3)	Bt toxin	–	Create pores in the epithelial cells of hindgut of insects
(4)	RNAi	–	Present in all prokaryotes as a method of cellular defense

170. The process of RNA interference initiates in the organism when ___ is present in its cell. Choose the option which **correctly** fills the blank.
(1) dsRNA (2) dsDNA
(3) ssRNA (4) ssDNA

171. How many recombinant therapeutics are presently being marketed in India?
(1) 30
(2) 27
(3) 12
(4) 10

172. What is **incorrect** about the human insulin produced by Eli Lilly, an American Company?
(1) Isolated genes coding for A and B chains from the human cell
(2) A and B chains were produced separately, extracted and combined by creating disulphide bonds
(3) Host cell used was *E.coli*
(4) This insulin can be used for the treatment of patients suffering from diabetes mellitus

173. ADA is crucial for all **except**
(1) Proper functioning of Immune system
(2) Enzyme replacement therapy for SCID
(3) Bone marrow transplantation
(4) Proliferation of lymphocytes

174. Select the **mismatch**.

(1)	Bt plants	–	Bt cotton, Bt corn, Bt rice, Bt soyabean
(2)	Transgenic models for human diseases	–	Cystic fibrosis, cancer, rheumatoid arthritis, Alzheimer's
(3)	Transgenic animals	–	Rabbits, fish, pigs, rats
(4)	Rosie	–	The first transgenic cow developed in 1987

175. An American company got patent rights on Basmati rice through the A office in the year B . Choose the option which **correctly** fill the blanks.

	A		B
(1)	US patent and trademark	–	1997
(2)	European patent	–	1997
(3)	Japan patent	–	1987
(4)	International patent	–	1987

176. The Texas based company, won a patent on Basmati rice line, they produced a new variety of Basmati rice by crossing
(1) Basmati rice × Semi-dwarf variety of rice
(2) IR-8 × Jaya
(3) IR-8 × Taichung Native-I
(4) Basmati rice × Sonalika

177. Biopiracy is the term used to refer
(1) A set of standards by which a community regulates its behaviour
(2) Decision regarding the validity of GM research
(3) Patents granted for biological entities
(4) Use of bio-resources without proper authorisation from the countries and people concerned

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178. Which of the following is an **incorrect** statement?

(1) Developing countries are rich in traditional knowledge and bio-resources

(2) There are an estimated 2,00,000 varieties of rice in India

(3) For chemical safety testing, transgenic animals are made less sensitive to toxic substances

(4) The Indian parliament has recently cleared the second amendment of the Indian Patents Bill

179. Which nematode causes root knot in the tobacco plant?

(1) *Agrobacterium tumefaciens*

(2) *Bacillus thuringiensis*

(3) *Meloidogyne incognitia*

(4) *Caenorhabditis elegans*

180. Bt toxin is an endotoxin which indicate that it is

(1) Released on lysis of bacteria

(2) Released as soon as it is formed inside the bacteria

(3) Formed during sporulation

(4) Produced by only some strains of *Bacillus thuringiensis*

❑ ❑ ❑

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